POWER PULSE

HVDC Basics Important Formulae

Notes by Balaji Sir

1 Six-pulse converter

1. $$V_{d0} = \frac{3\sqrt{2}}{\pi} V_{LL} \approx 1.35\, V_{LL}$$

2. $$V_d = V_{d0}\cos\alpha - \frac{3}{\pi} X_c I_d = V_{d0}\frac{\cos\alpha + \cos(\alpha+\mu)}{2}$$

rectifier

3. $$\cos\alpha - \cos(\alpha+\mu) = \frac{\sqrt{2}\, X_c I_d}{V_{LL}}$$

overlap angle μ

4. $$V_{di} = V_{d0}\cos\gamma - \frac{3}{\pi} X_c I_d, \qquad \gamma = \beta - \mu,\ \ \beta = 180^\circ - \alpha$$

inverter: α in 90°–180°

2 Link

5. $$I_d = \frac{V_{dr} - V_{di}}{R_d}, \quad P_r = V_{dr} I_d, \quad P_i = V_{di} I_d, \quad P_{loss} = I_d^2 R_d$$

6. $$P_{bipolar} = 2\, V_d I_d$$

7. 12-pulse: two 6-pulse bridges in series, $V_d = 2V_{d,\text{6-pulse}}$

3 Reactive power and harmonics

8. $$Q = P\tan\phi, \qquad \cos\phi \approx \frac{\cos\alpha + \cos(\alpha+\mu)}{2}$$

converters always absorb Q (≈ 50–60% of P)

9. $$I_{ac,rms} = \sqrt{\tfrac{2}{3}}\, I_d = 0.816 I_d, \quad I_1 = \frac{\sqrt{6}}{\pi} I_d = 0.78 I_d$$

10. Harmonics: AC $pk \pm 1$, DC pk $(p = 6:\ 5, 7, 11, 13;\ 6, 12; \quad p = 12:\ 11, 13;\ 12)$

4 Control and economics

11. Rectifier: constant current; inverter: constant extinction angle ($\gamma \approx 15^\circ$–18°); current margin ≈ 0.1 pu

12. $$d_{break-even} = \frac{\text{extra terminal cost of HVDC}}{\text{line cost saved per km}} \quad (\approx 600\text{–}800 \text{ km overhead})$$